TEXORA · LECTURE SHEET

Polymer Additives

The ingredients that turn a raw polymer into a usable material: plasticisers, stabilisers, delustrants, pigments, flame retardants, antistatic agents and processing aids — what each does, how much is used, and the textile examples that matter.

TE-2107 · POLYMER SCIENCE AND ENGINEERING · TOPIC 5 · LEVEL 2 TERM 1

1. Why additives are unavoidable

Almost no polymer is used pure. PVC without stabiliser degrades in its own processing heat; polyester without delustrant shines like glass; polypropylene without antioxidant crumbles in sunlight within a season. **Additives** are substances mixed into a polymer — usually in amounts from a fraction of a percent to tens of percent — to change its processing behaviour, its properties in service, or its appearance. Additive amounts are quoted either as **weight %** of the compound or, in rubber and PVC practice, as **phr** (parts per hundred of resin): 40 phr means 40 parts additive to 100 parts polymer.

WORKED EXAMPLE — PHR AND PERCENTAGE

A flexible PVC compound carries 40 phr of plasticiser DOP. Percentage in the finished compound = 40 ÷ (100 + 40) × 100 = **28.6 %**.

A 500 kg batch of semi-dull PET chips contains 0.30 % TiO_2: delustrant mass = 500 × 0.0030 = **1.5 kg**. The same arithmetic at full-dull level (2.0 %) would need 10 kg — which is why dullness is specified to two decimal places in purchase contracts.

2. Plasticisers

Plasticisers are low-volatility liquids (classically phthalate esters such as DOP/DEHP) whose molecules insert themselves between polymer chains, push them apart, weaken the inter-chain forces and lower T_g. A glassy polymer becomes soft and flexible. PVC is the great user: rigid PVC (uPVC, no plasticiser) makes pipes; with 30–50 phr plasticiser it becomes the flexible film of rainwear, synthetic leather and cable covering. Plasticiser can slowly migrate out and evaporate — the reason old PVC articles stiffen and crack — and health rules now restrict several phthalates in consumer goods, pushing the industry toward alternatives such as DINP, DOTP and citrate esters. In fibres, plasticisation appears in a gentler form: water itself plasticises nylon and viscose (lowering T_g and stiffness when wet), and carriers plasticise PET during dyeing.

3. Stabilisers — defending the chain

3.1 Heat stabilisers

PVC begins to release HCl at processing temperatures, and the released HCl catalyses further degradation (an unzipping "autocatalytic" attack that yellows and embrittles the product). Heat stabilisers — historically lead salts, now mainly calcium–zinc and organotin systems — scavenge HCl and interrupt the cycle. Polyester and nylon melts carry thermal stabilisers (phosphorus compounds in PET) to limit chain scission during the minutes spent at 280 °C.

3.2 Antioxidants

Oxygen attacks hot polymer during processing and warm polymer in service, through a free-radical chain reaction that cuts chains. **Primary antioxidants** (hindered phenols such as Irganox types) donate hydrogen to kill the radicals; **secondary antioxidants** (phosphites, thioesters) decompose the hydroperoxide intermediates. Polypropylene, whose tertiary carbons are especially vulnerable, is unusable outdoors without them.

3.3 UV stabilisers and screens

Sunlight's ultraviolet breaks bonds directly (photo-degradation — Topic 6). Defences: **UV absorbers** (benzophenones, benzotriazoles) that soak up UV and release it as heat; **HALS** (hindered amine light stabilisers) that trap the radicals UV creates; and simple **screens** — carbon black and TiO_2 particles that block light physically. Nylon and polyester lose strength steadily in sunlight; acrylic's inherent UV resistance is a chain-structure advantage, not an additive.

4. Additives that manage appearance

4.1 Delustrants

Melt-spun filaments are naturally glassy-bright. **Titanium dioxide (TiO_2)** particles dispersed in the polymer scatter light inside the filament and kill the shine:

GRADE	TIO_2 CONTENT	APPEARANCE
Bright	0 % (or trace)	Glassy lustre — hosiery, linings
Semi-dull	≈ 0.3 %	Soft sheen — most apparel filament
Full dull	≈ 1.5–2.5 %	Matt, cotton-like — suiting, sportswear

TiO_2 is a double-edged additive: it also mildly photocatalyses surface degradation and abrades spinnerets and guides, so its level is a compromise, not a maximum.

4.2 Pigments and colorants

Insoluble **pigments** (inorganic: iron oxides, carbon black; organic: phthalocyanine blue/green) are dispersed in the melt for dope-dyed fibres — the only colouration route for polypropylene and a fast-growing route for polyester (water-free dyeing with excellent fastness). Soluble **dyes** can be added to some plastics for transparent colours but are rarely used inside fibres because they can migrate and bleed. **Optical brighteners** absorb UV and re-emit blue light, cancelling the yellow cast of "white" articles.

5. Functional additives

ADDITIVE	FUNCTION	MECHANISM / EXAMPLE
Flame retardants	Slow ignition and flame spread	Halogen compounds release flame-quenching radicals; phosphorus compounds char the surface; aluminium trihydrate releases water. Used in contract furnishings, children's nightwear finishes, protective clothing
Antistatic agents	Prevent static charge build-up	Hygroscopic compounds (ethoxylated amines, quaternary salts) draw a moisture film that leaks charge away; spin finishes carry them for synthetic fibres
Fillers	Cheapness, stiffness, dimensional stability	$CaCO_3$, talc, glass fibre in plastics; raise modulus but lower impact strength and add weight
Lubricants / processing aids	Ease flow and release from moulds and dies	Stearates, waxes; reduce melt viscosity and die drool in extrusion and spinning
Antimicrobials	Suppress bacterial growth and odour	Silver ions, zinc compounds or organic biocides built into hospital textiles, socks, sportswear; must survive washing
Cross-linking (curing) agents	Convert thermoplastic chains into networks	Sulfur in rubber vulcanisation; peroxides; resin finishes (DMDHEU) cross-link cotton cellulose for easy-care fabrics
Blowing agents	Create foam structure	Chemical agents decompose to N_2/CO_2 on heating — expanded PVC and PS foams, foam-backed coatings
Spin finish (fibre-specific)	Lubricate and protect filaments through spinning, drawing and textile processing	An emulsion of lubricant + antistat + emulsifier applied at fractions of a percent on filament weight; removed by scouring before dyeing

6. Rules for choosing additives

- **Compatibility** — the additive must stay dispersed; incompatibility shows as blooming (additive sweating to the surface), specks in filaments and spinneret blockage.
- **Permanence** — it must resist evaporation, extraction by water and detergents, and migration into neighbouring materials for the article's whole life.
- **Process survival** — a fibre additive must withstand 280–290 °C melt temperatures without decomposing or discolouring.
- **No property sabotage** — every additive trades something away: plasticiser lowers strength and heat resistance; TiO_2 abrades machinery; fillers embrittle.
- **Regulation** — skin-contact textiles face strict limits (phthalates, certain flame retardants, formaldehyde-releasing finishes) under rules such as OEKO-TEX and REACH.

Glossary

phr — parts of additive per hundred parts of resin by weight. **Plasticiser** — a liquid additive that separates chains and lowers T_g, making the polymer flexible. **HALS** — hindered amine light stabiliser, a radical-trapping UV protector. **Delustrant** — a light-scattering pigment (TiO_2) that removes filament shine. **Blooming** — migration of an incompatible additive to the surface as a visible film. **Spin finish** — the temporary lubricant/antistat coating applied to filaments for processing.

Key points

• No commercial polymer is used pure; additives fix processing, durability and appearance at 0.1 % to 50 % levels.
• Plasticisers work by separating chains and lowering Tg — PVC ranges from rigid pipe to flexible film on plasticiser content alone.
• Stabiliser families map onto enemies: heat stabilisers vs processing heat, antioxidants vs oxygen, UV absorbers/HALS/screens vs sunlight.
• TiO_2 at ≈ 0.3 % (semi-dull) to ≈ 2 % (full dull) sets the lustre grade of synthetic filaments.
• Every additive is a trade: permanence, compatibility, process survival and regulation filter the choice.

Practice

1. A compound is made from 100 kg PP, 1.5 kg antioxidant package and 25 kg calcium carbonate filler. Express the filler in phr and state the antioxidant level in %.

2. An old PVC raincoat has become stiff and cracked. Explain the mechanism, naming the additive involved.
3. Why does a polyester producer choose 0.3 % rather than 2 % TiO_2 for a hosiery filament, even though customers want a matt look? Give two reasons.
4. Map each enemy to its defensive additive family: (a) HCl released during PVC processing; (b) radical attack by oxygen in hot PP; (c) sunlight on outdoor nylon webbing.
5. A white polyester fabric looks yellowish although it is clean. Which additive class corrects the appearance, and how does it work?

Answer key

1. Filler = 25 phr (25 parts per 100 parts PP). Antioxidant = 1.5 ÷ 126.5 × 100 ≈ **1.19 %** of the total compound (1.5 % relative to polymer alone).
2. Plasticiser has migrated out and evaporated over the years; with the chains no longer separated, T_g of the PVC rises back toward room temperature and beyond, so the compound returns to its natural glassy, brittle state and cracks on flexing.
3. (i) Abrasion and photochemistry: TiO_2 particles wear spinnerets, guides and needles and photocatalyse surface degradation, shortening equipment and fabric life. (ii) Fine hosiery filaments are thin; a heavy particle load raises filament breaks during spinning and drawing and can show as specks, while 0.3 % already removes the objectionable glassy shine.
4. (a) Heat stabilisers (Ca–Zn or organotin HCl scavengers). (b) Antioxidants — hindered phenol (primary) with phosphite (secondary). (c) UV absorbers/HALS, ideally with a screening pigment; carbon black is the strongest screen where colour allows.
5. Optical brighteners (fluorescent whitening agents). They absorb invisible ultraviolet and re-emit it as blue visible light; the added blue cancels the yellow reflectance, so the fabric appears a brighter, bluer white.
